SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4 **Total Marks:40**

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

SOLUTIONS:

Case Study 1: Video Conferencing

The company experiences 8% packet loss, 180 ms jitter, distorted voice, frozen video, and delayed communication during peak hours. The case specifically asks whether TCP/UDP or application-level buffering is responsible.

1.Analyse whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.

The problem is primarily related to the transport protocol and network conditions, with application-level buffering contributing to the issue.

Real-time video conferencing normally uses UDP because it provides low latency and does not wait for retransmission of lost packets. However, the observed 8% packet loss and 180 ms jitter are severe enough to affect voice and video quality.

If TCP is used, lost packets are retransmitted and packets may arrive in order, but this can introduce additional delay and head-of-line blocking. Therefore, TCP is not ideal for highly interactive real-time communication. The application layer also plays an important role. If the jitter buffer is too small, packets arriving at different times may cause audio/video gaps. If it is too large, it increases end-to-end delay.

Hence, the main issue is network/transport performance, amplified by the application's buffering strategy.

2. Justify your answer based on the communication requirements of real-time multimedia applications.

Real-time multimedia applications require:

- Low latency for natural conversation.
- Low jitter for smooth audio and video.
- Low packet loss to prevent distortion and freezing.
- Continuous transmission rather than perfect delivery of every packet.
- Fast adaptation to changing bandwidth.

UDP is generally preferred because it avoids retransmission delays and provides faster delivery. Small amounts of packet loss can be handled using techniques such as Forward Error Correction (FEC), packet-loss concealment, and adaptive bitrate.

In this case, 180 ms jitter and 8% packet loss directly explain the frozen frames, distorted voice, and delayed communication.

4.Recommend suitable protocol-level or application-level improvements.

The following improvements are recommended:

1. Use UDP-based real-time transport instead of TCP where appropriate.
2. Apply adaptive bitrate streaming so video quality automatically decreases when bandwidth falls.
3. Use an adaptive jitter buffer to compensate for variable packet arrival times.
4. Implement Forward Error Correction (FEC) to recover from some packet losses without retransmission.
5. Apply QoS prioritization to give real-time voice/video traffic higher priority.
6. Use congestion-control mechanisms suitable for real-time traffic.
7. Reduce unnecessary buffering to maintain low end-to-end latency.
8. Increase available bandwidth or optimize network capacity during peak hours.

Conclusion: Combining UDP-based transport, QoS, adaptive bitrate, and intelligent buffering would significantly improve conferencing quality.

Case Study 2: Global Web Services and DNS

The company experiences approximately **800 ms DNS lookup time** and wants to reduce it below **50 ms** while maintaining availability during server failures.

1.Analyse the reasons behind the high DNS resolution time.

The major reasons for the 800 ms DNS resolution time are:

- DNS servers may be geographically far from users.
- Requests may travel through multiple network hops.
- DNS caching may be ineffective because of inappropriate TTL values.
- Recursive DNS resolvers may need to contact authoritative servers repeatedly.
- Heavy traffic may overload DNS servers.
- Lack of geographically distributed DNS infrastructure increases round-trip time.
- DNS prefetching may not be used by the application or browser.
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Therefore, a centralized or poorly distributed DNS architecture can significantly increase resolution latency.

2. Propose a suitable DNS architecture to reduce lookup time below 50 ms.

A **distributed Anycast DNS architecture** should be implemented.

Proposed architecture:
Users → Nearest Anycast DNS Resolver → Regional DNS Cache → Authoritative DNS Servers

The company should deploy DNS servers at multiple geographic locations. All servers can advertise the same Anycast IP address. Internet routing will direct users to a nearby DNS location.

The architecture should include:

- **Anycast DNS** – directs users to the nearest available DNS server.
- **DNS caching** – stores frequently requested records closer to users.
- **Optimized TTL values** – keeps records cached for an appropriate period.
- **DNS prefetching** – resolves important domain names before they are actually needed.
- **Multiple authoritative servers** – provides redundancy.
- **Health monitoring and automatic failover** – removes failed servers from service.
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With geographically distributed caching and Anycast routing, DNS lookup latency can potentially be reduced to **below 50 ms**, depending on network conditions.

4.Evaluate advantages and limitations.

Aspect	Advantages	Limitations
Performance	Lower geographic latency	Requires multiple DNS locations
Scalability	Handles large global traffic	Higher infrastructure cost
Availability	Failure of one server does not stop DNS	Configuration is more complex
Anycast	Routes users to nearby servers	Routing problems can affect traffic
Caching	Reduces repeated DNS queries	Stale records may remain until TTL expires
Prefetching	Reduces lookup delay	Can generate unnecessary queries

Conclusion: A globally distributed Anycast DNS architecture with caching, appropriate TTLs, and prefetching provides better performance, scalability, and fault tolerance.

Case Study 3: Cloud Service Provider

The cloud provider serves millions of users, but all DNS requests are directed to **one primary DNS server**, resulting in approximately **800 ms lookup time** during promotional events.

4.Examine the impact of centralized DNS architecture on application performance.

A centralized DNS architecture creates a **single point of failure and performance bottleneck**.

When millions of users send DNS queries to one server:

- The DNS server becomes overloaded.
- Query processing time increases.
- Users experience longer DNS resolution delays.
- Application startup and webpage loading become slower.
- Network traffic toward the central server increases.
- Failure of the primary DNS server can make services unreachable.

For example, the case reports approximately **800 ms DNS lookup time**, which adds significant delay before users can connect to the cloud application.

Therefore, centralized DNS directly reduces application responsiveness and availability.

5.Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.

A suitable solution is:
Global Users
↓
Anycast IP
↓
Nearest Regional DNS Server
↓
Local/Recursive Cache
↓
Multiple Authoritative DNS Servers

The company should deploy DNS servers in major geographical regions such as Asia, Europe, and North America.

Each DNS location should:

- Advertise the same Anycast IP.
- Maintain cached DNS records.
- Use health monitoring.
- Automatically remove failed servers from routing.
- Have redundant authoritative DNS servers.
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This ensures that users are directed to a nearby operational DNS server instead of a single global server.

6.Analyse how TTL values, Anycast routing, and DNS caching influence system performance and fault tolerance.

TTL (Time To Live):
TTL determines how long a DNS record can remain cached.

- Higher TTL → fewer DNS queries and better performance.
- Lower TTL → faster propagation of changes but more DNS queries.

Therefore, TTL should be optimized according to how frequently DNS records change.

Anycast Routing:
Anycast allows multiple DNS servers to use the same IP address. Routing directs users toward an appropriate nearby server.
Benefits include:

- Lower latency.
- Load distribution.
- Improved availability.
- Automatic routing around some failures.

DNS Caching:
Caching stores previously resolved DNS records.
It:

- Reduces repeated DNS queries.
- Decreases DNS server load.
- Reduces lookup latency.
- Improves application response time.
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However, outdated cached records may remain until their TTL expires.

Conclusion: Combining optimized TTL values, Anycast routing, and distributed caching provides high performance and fault tolerance.

Case Study 4: Electronic Stock Trading

The financial institution experiences an increase in order acknowledgment latency from **1 ms to 40 ms**. Engineers observe increased TCP retransmissions, longer connection establishment, and excessive buffering. The case specifically asks about **flow control, Nagle's algorithm, and SYN queue limitations**.

1. Analyze three TCP-related factors responsible for the latency increase.

A. TCP Flow Control
TCP uses the receiver's advertised window to prevent the sender from transmitting more data than the receiver can handle.
During heavy trading activity, the receive buffer may become full. The sender must then slow down or wait for the receiver to advertise more window space.

This can cause:

- Lower throughput.
- Increased waiting time.
- Delayed acknowledgments.
- Higher transaction latency.

B. Nagle's Algorithm
Nagle's algorithm combines small TCP packets into larger segments to reduce network overhead.
This is useful for normal applications, but stock trading frequently involves **small, latency-sensitive messages**. Nagle's algorithm may wait for an acknowledgment before sending another small segment, creating additional delay.

Therefore, it can be undesirable for high-frequency trading systems.

C. SYN Queue Limitations
When clients establish TCP connections, the server maintains structures for partially completed connections. During market opening, thousands of connection attempts may arrive simultaneously. If the SYN queue becomes full:

- New connections may be delayed.
- SYN packets may be retransmitted.
- Connection establishment takes longer.
- Some connection attempts may fail.

This directly increases transaction latency.

2. Explain how each factor affects transaction processing in high-frequency systems.

TCP Factor	Effect on High-Frequency Transactions
Flow Control	Limits transmission when receiver buffers are full
Nagle's Algorithm	Can delay small trading messages
SYN Queue	Overload delays new TCP connections
Retransmissions	Increase delivery time
Buffering	Adds waiting time before packets are transmitted

In high-frequency trading, even a few milliseconds can be significant. Increasing latency from **1 ms to 40 ms** can delay order acknowledgments and trade confirmations, potentially affecting the timing of transactions.

3. Recommend appropriate TCP configuration changes.

The following changes are recommended:

1. **Disable Nagle's algorithm** using TCP_NODELAY for latency-sensitive trading connections.
2. Increase and properly tune **TCP receive/send buffers** to prevent unnecessary flow-control blocking.

3. Increase the server's **SYN backlog/queue capacity** to handle bursts of connection requests.
4. Use **persistent TCP connections** wherever possible to avoid repeated TCP handshakes.
5. Monitor and reduce **TCP retransmissions** by improving network reliability.
6. Use appropriate **TCP congestion-control settings** for the network environment.
7. Reduce excessive application and kernel buffering where low latency is more important than throughput.
8. Use dedicated, high-performance network infrastructure for trading traffic.

Conclusion: Disabling Nagle's algorithm, tuning TCP buffers and SYN queues, maintaining persistent connections, and reducing retransmissions can significantly lower order acknowledgment latency while maintaining reliable delivery.
